CASB statistical analysis and paper results

A. Statistical methodology

Analysis uses saved evaluation records only; no model was trained. Each method/classifier/dataset score averages all ten supplied fold/seed evaluations. Incomplete groups are withheld. Primary tests keep classifiers separate. G-mean and minority F1 are primary metrics; precision and recall are secondary. Scores retain full precision; presentation tables show percentages. Numerical tie tolerance is 1e-12.

Two-sided Wilcoxon signed-rank tests compare each fixed CASB variant against every compatible baseline. Zeros are discarded; differences are rounded to 12 decimals for ranks, and the tie-corrected asymptotic p-value is used. The null concerns a paired-difference distribution symmetric about zero; asymmetry and related datasets limit interpretation. Rank-biserial effects, raw and adjusted p-values, exact sign-test sensitivity, paired units, and win/tie/loss counts are saved.

Holm families are fixed within classifier and analysis population across all four metrics: 4 variants x 7 baselines x 4 metrics = 112 for SVM/KNN; 4 x 5 x 4 = 80 for RF/LR/GaussianNB. The baseline set includes no resampling. RUS and SMOTE-ENN are unavailable under the RF/LR/GaussianNB protocol. The original five-baseline 80-test correction is also supplied in p_holm_core80, as a separate scope rather than a favourable replacement. No primary correction pools classifiers. Sensitivity populations have their own identical-size correction families.

Friedman tests use the same strict complete datasets and method set within each classifier, with Holm correction over four metrics. Significant omnibus tests establish differences somewhere among methods. Post hoc all-pairs Wilcoxon-Holm results use the same complete cohort, with a separate family covering all method pairs and four metrics for each classifier. Primary fixed-variant tests use all valid pairs, so comparisons without ADASYN can use 44 datasets while ADASYN uses 40.

Extraction and protocol audit

All 44 original dataset notebooks were read and verified executed. Original SVM/KNN current records: 7,920, including 26 failed evaluations. Expanded current RF/LR/GaussianNB records and compatible RUS/SMOTE-ENN evidence yield 21,560 unique saved evaluations. Successful: 21,495; failed: 65. All failures remain visible in the executed notebook.

Training-only preprocessing and a 25% stratified inner holdout selected ratios 0.5/0.75/1.0 by (G-mean + F1)/2, ties preferring the lower ratio. Seeds were 42/137, using the supplied five train/test folds. The two seeds reuse the same supplied folds and are averaged, never counted as independent datasets. Legacy RUS/ENN compatibility was checked by equality of every shared current/legacy record (scores, confusion matrices, ratios and errors). Historical protocols outside these metadata sections were not merged. Abalone uses training-fitted one-hot inputs; interpolation may yield fractional dummy values.

RF and LR were selected for presentation after examining results; GaussianNB is reported separately. None of these are independent confirmatory evaluations. Overlap-simulation records are excluded from the real-dataset statistical population.

classifier	method	dataset	failed
gnb	adasyn	ecoli-0-1-3-7_vs_2-6	1
gnb	adasyn	iris0	5
gnb	adasyn	shuttle-c0-vs-c4	4
gnb	adasyn	shuttle-c2-vs-c4	3
knn	adasyn	ecoli-0-1-3-7_vs_2-6	1
knn	adasyn	iris0	5
knn	adasyn	shuttle-c0-vs-c4	4

classifier	method	dataset	failed
knn	adasyn	shuttle-c2-vs-c4	3
lr	adasyn	ecoli-0-1-3-7_vs_2-6	1
lr	adasyn	iris0	5
lr	adasyn	shuttle-c0-vs-c4	4
lr	adasyn	shuttle-c2-vs-c4	3
rf	adasyn	ecoli-0-1-3-7_vs_2-6	1
rf	adasyn	iris0	5
rf	adasyn	shuttle-c0-vs-c4	4
rf	adasyn	shuttle-c2-vs-c4	3
svm	adasyn	ecoli-0-1-3-7_vs_2-6	1
svm	adasyn	iris0	5
svm	adasyn	shuttle-c0-vs-c4	4
svm	adasyn	shuttle-c2-vs-c4	3

B. Overall performance

SVM: 40 strict matched datasets

method	gmean	f1	precision	recall
adasyn	81.499	64.737	59.979	76.397
bsmote	79.669	65.610	61.503	74.052
casb	80.484	67.276	65.240	74.435
casb_edited	82.638	66.102	61.197	78.510
casb_relaxed	81.376	67.108	64.257	75.203
casb_ros	80.964	**67.444**	65.313	75.003
original	59.194	54.653	**66.558**	49.652
ros	82.620	66.951	63.344	76.906
rus	80.546	61.065	56.011	77.551
smote	81.619	66.608	63.378	75.738
smoteenn	**83.165**	65.165	60.382	**79.324**

KNN: 40 strict matched datasets

method	gmean	f1	precision	recall
adasyn	83.200	63.157	55.930	80.001

method	gmean	f1	precision	recall
bsmote	81.884	64.685	58.208	77.364
casb	82.098	**65.355**	59.939	76.244
casb_edited	**83.404**	63.408	56.056	80.618
casb_relaxed	82.399	64.642	59.108	77.038
casb_ros	81.044	64.436	59.180	74.908
original	65.816	60.354	**70.035**	56.225
ros	81.691	64.396	58.588	76.221
rus	82.084	59.473	53.010	79.783
smote	82.702	64.554	59.028	77.816
smoteenn	83.263	62.550	55.397	**81.535**

RF: 40 strict matched datasets

method	gmean	f1	precision	recall
adasyn	77.289	66.828	66.513	69.572
bsmote	73.983	65.772	68.228	66.452
casb	75.499	67.594	71.320	66.994
casb_edited	**79.320**	67.619	64.907	**73.502**
casb_relaxed	76.857	67.855	69.556	68.665
casb_ros	75.171	67.843	**72.014**	67.056
original	64.516	59.401	70.564	54.879
ros	75.239	67.592	71.133	67.040
smote	77.153	**67.907**	69.424	68.941

LR: 40 strict matched datasets

method	gmean	f1	precision	recall
adasyn	84.084	62.879	55.337	80.997
bsmote	82.540	63.157	56.079	78.533
casb	83.235	63.998	58.205	78.232
casb_edited	**84.284**	63.207	55.311	**82.250**
casb_relaxed	83.609	64.027	57.756	79.477
casb_ros	83.510	**64.112**	58.122	78.832
original	59.700	53.745	**67.216**	48.167
ros	84.139	63.734	57.317	80.386

method	gmean	f1	precision	recall
smote	83.943	63.749	57.216	79.978

GNB: 40 strict matched datasets

method	gmean	f1	precision	recall
adasyn	66.438	45.292	37.863	**86.790**
bsmote	67.494	45.947	38.545	84.196
casb	66.023	45.111	38.405	84.788
casb_edited	**68.310**	**46.971**	39.497	85.216
casb_relaxed	67.162	45.671	38.779	84.762
casb_ros	64.335	44.534	37.806	85.097
original	63.339	44.964	**39.531**	81.423
ros	63.675	44.508	37.692	85.459
smote	67.699	46.142	39.193	85.470

Best observed variant averages versus all available baselines

Each row selects the highest collection-wide variant average for a metric. This differs from selecting a different variant within every dataset. Both summaries are descriptive.

classifier	metric	best_variant	family_average_percent	best_baseline	baseline_average_percent	difference_pp
svm	gmean	casb_edited	82.638	smoteenn	83.165	-0.526
svm	f1	casb_ros	67.444	ros	66.951	0.494
svm	precision	casb_ros	65.313	original	66.558	-1.245
svm	recall	casb_edited	78.510	smoteenn	79.324	-0.815
knn	gmean	casb_edited	83.404	smoteenn	83.263	0.140
knn	f1	casb	65.355	bsmote	64.685	0.670
knn	precision	casb	59.939	original	70.035	-10.096
knn	recall	casb_edited	80.618	smoteenn	81.535	-0.917
rf	gmean	casb_edited	79.320	adasyn	77.289	2.031
rf	f1	casb_relaxed	67.855	smote	67.907	-0.052
rf	precision	casb_ros	72.014	ros	71.133	0.881
rf	recall	casb_edited	73.502	adasyn	69.572	3.930
lr	gmean	casb_edited	84.284	ros	84.139	0.145
lr	f1	casb_ros	64.112	smote	63.749	0.363
lr	precision	casb	58.205	original	67.216	-9.011
lr	recall	casb_edited	82.250	adasyn	80.997	1.253

classifier	metric	best_variant	family_average_percent	best_baseline	baseline_average_percent	difference_pp
gnb	gmean	casb_edited	68.310	smote	67.699	0.612
gnb	f1	casb_edited	46.971	smote	46.142	0.829
gnb	precision	casb_edited	39.497	original	39.531	-0.035
gnb	recall	casb_edited	85.216	adasyn	86.790	-1.573

C. Friedman results

classifier	metric	n_datasets	n_methods	statistic	p_raw	p_holm_four_metrics	significant
svm	gmean	40	11	88.4667	1.07802e-14	2.15605e-14	True
svm	f1	40	11	69.4964	5.54586e-11	5.54586e-11	True
svm	precision	40	11	104.549	6.67251e-18	2.00175e-17	True
svm	recall	40	11	155.457	2.8028e-28	1.12112e-27	True
knn	gmean	40	11	94.0589	8.36272e-16	1.67254e-15	True
knn	f1	40	11	44.0129	3.27417e-06	3.27417e-06	True
knn	precision	40	11	121.653	2.33635e-21	7.00905e-21	True
knn	recall	40	11	174.373	3.44456e-32	1.37782e-31	True
rf	gmean	40	9	109.298	5.30638e-20	1.59191e-19	True
rf	f1	40	9	51.1262	2.48121e-08	2.48121e-08	True
rf	precision	40	9	87.5578	1.45476e-15	2.90952e-15	True
rf	recall	40	9	137.455	8.02328e-26	3.20931e-25	True
lr	gmean	40	9	95.7329	3.17169e-17	9.51507e-17	True
lr	f1	40	9	34.1944	3.74597e-05	3.74597e-05	True
lr	precision	40	9	86.4502	2.43837e-15	4.87674e-15	True
lr	recall	40	9	146.575	1.01521e-27	4.06083e-27	True
gnb	gmean	40	9	45.7792	2.61809e-07	1.04724e-06	True
gnb	f1	40	9	34.6183	3.13761e-05	6.27522e-05	True
gnb	precision	40	9	35.8419	1.87694e-05	5.63083e-05	True
gnb	recall	40	9	29.3549	0.000274679	0.000274679	True

A significant Friedman result does not identify a superior CASB variant. Mean ranks and the complete method/dataset lists are stored in friedman_results.csv. All-pairs post hoc results are rows labelled friedman_posthoc in wilcoxon_holm_results.csv.

D. Wilcoxon results

classifier	CASB higher	CASB lower	not significant
gnb	3	1	76

classifier	CASB higher	CASB lower	not significant
knn	18	13	81
lr	14	5	61
rf	21	4	55
svm	27	4	81

All significant improvements and decreases (scores/differences below remain fractions, not percentages):

classifier	metric	method	baseline	n	mean_difference	rank_biserial	p_raw	p_holm	direction
svm	gmean	casb	original	44	0.193775	0.990244	4.83792e-08	5.32171e-06	CASB higher
svm	gmean	casb_relaxed	original	44	0.201878	0.990244	4.83792e-08	5.32171e-06	CASB higher
svm	gmean	casb_edited	original	44	0.213274	0.984496	2.73073e-08	3.05842e-06	CASB higher
svm	gmean	casb_ros	original	44	0.198098	0.990244	4.83792e-08	5.32171e-06	CASB higher
svm	f1	casb	original	44	0.114986	0.797561	1.10618e-05	0.00107299	CASB higher
svm	f1	casb	rus	44	0.0596943	0.816168	4.07343e-06	0.000407343	CASB higher
svm	f1	casb_relaxed	original	44	0.113371	0.858537	2.23046e-06	0.000227507	CASB higher
svm	f1	casb_relaxed	rus	44	0.058079	0.796235	6.9548e-06	0.00068157	CASB higher
svm	f1	casb_relaxed	smoteenn	44	0.0215023	0.629268	0.000524631	0.0445936	CASB higher
svm	f1	casb_edited	original	44	0.101802	0.703212	7.18988e-05	0.00647089	CASB higher
svm	f1	casb_edited	rus	44	0.04651	0.640592	0.000253487	0.0220534	CASB higher
svm	f1	casb_ros	original	44	0.116467	0.795122	1.17669e-05	0.00112962	CASB higher
svm	f1	casb_ros	rus	44	0.0611749	0.769657	1.39261e-05	0.00132298	CASB higher
svm	precision	casb	adasyn	40	0.0526161	0.660661	0.000547592	0.0459977	CASB higher
svm	precision	casb	rus	44	0.0890054	0.831672	2.66443e-06	0.000269107	CASB higher
svm	precision	casb	smoteenn	44	0.0495598	0.763869	5.10717e-05	0.00464752	CASB higher
svm	precision	casb_relaxed	adasyn	40	0.0427803	0.666667	0.000582531	0.0477676	CASB higher
svm	precision	casb_relaxed	rus	44	0.0799019	0.847176	1.72998e-06	0.000179918	CASB higher
svm	precision	casb_relaxed	smoteenn	44	0.0404563	0.72973	0.000109	0.00970097	CASB higher
svm	precision	casb_edited	smote	44	-0.0228569	-0.797436	1.4248e-05	0.00133931	CASB lower
svm	precision	casb_edited	ros	44	-0.0144598	-0.633333	0.000567003	0.0470612	CASB lower
svm	precision	casb_ros	rus	44	0.089662	0.802879	5.82681e-06	0.000576854	CASB higher
svm	precision	casb_ros	smoteenn	44	0.0502164	0.672831	0.000359845	0.0309466	CASB higher
svm	recall	casb	original	44	0.225776	1	5.24943e-08	5.61689e-06	CASB higher
svm	recall	casb	smoteenn	44	-0.044354	-0.810084	3.77951e-05	0.00347715	CASB lower
svm	recall	casb_relaxed	original	44	0.232754	1	5.25139e-08	5.61689e-06	CASB higher
svm	recall	casb_relaxed	smoteenn	44	-0.0373753	-0.832442	3.01204e-05	0.00280119	CASB lower
svm	recall	casb_edited	original	44	0.262814	1	3.56686e-08	3.95922e-06	CASB higher

classifier	metric	method	baseline	n	mean_difference	rank_biserial	p_raw	p_holm	direction
svm	recall	casb_edited	smote	44	0.0252893	0.932773	2.07906e-06	0.000214143	CASB higher
svm	recall	casb_edited	bsmote	44	0.0409977	0.756303	0.000119609	0.0105256	CASB higher
svm	recall	casb_ros	original	44	0.230843	1	5.25334e-08	5.61689e-06	CASB higher
knn	gmean	casb	original	44	0.147531	0.934959	1.83085e-07	1.95901e-05	CASB higher
knn	gmean	casb_relaxed	original	44	0.150482	0.934959	1.83085e-07	1.95901e-05	CASB higher
knn	gmean	casb_edited	original	44	0.159353	0.934959	1.83085e-07	1.95901e-05	CASB higher
knn	gmean	casb_ros	original	44	0.138051	0.941928	1.48335e-07	1.60202e-05	CASB higher
knn	gmean	casb_ros	adasyn	40	-0.021559	-0.769231	2.8329e-05	0.00254961	CASB lower
knn	f1	casb	smoteenn	44	0.0254367	0.628339	0.000456203	0.0378648	CASB higher
knn	f1	casb_relaxed	smoteenn	44	0.020832	0.723577	5.42479e-05	0.00477381	CASB higher
knn	precision	casb	original	44	-0.0983832	-0.62439	0.000579628	0.0475295	CASB lower
knn	precision	casb	adasyn	40	0.040087	0.889744	1.2829e-06	0.000130856	CASB higher
knn	precision	casb	smoteenn	44	0.0415267	0.788618	1.08569e-05	0.00106398	CASB higher
knn	precision	casb_relaxed	original	44	-0.104287	-0.678049	0.000186465	0.0158496	CASB lower
knn	precision	casb_relaxed	adasyn	40	0.0317751	0.797436	1.42446e-05	0.00133899	CASB higher
knn	precision	casb_relaxed	smoteenn	44	0.0356226	0.89547	5.86995e-07	6.04605e-05	CASB higher
knn	precision	casb_edited	original	44	-0.13386	-0.773171	2.03569e-05	0.00187284	CASB lower
knn	precision	casb_edited	smote	44	-0.0278921	-0.865854	1.82713e-06	0.00018454	CASB lower
knn	precision	casb_edited	ros	44	-0.0237374	-0.730769	6.9739e-05	0.00606729	CASB lower
knn	precision	casb_ros	original	44	-0.104742	-0.659756	0.000276999	0.0232679	CASB lower
knn	precision	casb_ros	adasyn	40	0.0324955	0.764103	3.20219e-05	0.00284995	CASB higher
knn	precision	casb_ros	smoteenn	44	0.0351682	0.788618	1.08569e-05	0.00106398	CASB higher
knn	recall	casb	original	44	0.181989	1	7.7367e-08	8.50704e-06	CASB higher
knn	recall	casb	adasyn	40	-0.0375658	-0.887302	4.67789e-06	0.000467789	CASB lower
knn	recall	casb	smoteenn	44	-0.0481027	-0.792173	2.07313e-05	0.00188655	CASB lower
knn	recall	casb_relaxed	original	44	0.189206	1	7.73367e-08	8.50704e-06	CASB higher
knn	recall	casb_relaxed	adasyn	40	-0.0296277	-0.868254	7.45394e-06	0.00073794	CASB lower
knn	recall	casb_relaxed	smoteenn	44	-0.0408862	-0.827881	1.12807e-05	0.00108294	CASB lower
knn	recall	casb_edited	original	44	0.221753	1	5.2553e-08	5.87937e-06	CASB higher
knn	recall	casb_edited	smote	44	0.0254738	0.759664	0.000111341	0.00957531	CASB higher
knn	recall	casb_edited	ros	44	0.0399743	0.826984	1.98072e-05	0.00184207	CASB higher
knn	recall	casb_ros	original	44	0.169847	1	5.24943e-08	5.87937e-06	CASB higher
knn	recall	casb_ros	adasyn	40	-0.0509222	-0.978979	3.0199e-07	3.1407e-05	CASB lower
knn	recall	casb_ros	smoteenn	44	-0.0602449	-0.826458	1.16887e-05	0.00111043	CASB lower
rf	gmean	casb	original	44	0.11412	0.95122	1.11782e-07	8.16009e-06	CASB higher

classifier	metric	method	baseline	n	mean_difference	rank_biserial	p_raw	p_holm	direction
rf	gmean	casb_relaxed	original	44	0.126464	0.958188	9.0256e-08	6.7692e-06	CASB higher
rf	gmean	casb_edited	original	44	0.148854	0.974448	5.44803e-08	4.19498e-06	CASB higher
rf	gmean	casb_edited	bsmote	44	0.0485127	0.753846	4.08234e-05	0.00253105	CASB higher
rf	gmean	casb_edited	ros	44	0.0393673	0.746341	3.90488e-05	0.00246007	CASB higher
rf	gmean	casb_ros	original	44	0.11114	0.976771	5.06567e-08	4.00188e-06	CASB higher
rf	f1	casb	original	44	0.0881178	0.87921	9.35517e-07	6.6131e-05	CASB higher
rf	f1	casb_relaxed	original	44	0.0904923	0.860627	1.57819e-06	0.000108895	CASB higher
rf	f1	casb_edited	original	44	0.0883484	0.839721	2.80717e-06	0.000190888	CASB higher
rf	f1	casb_ros	original	44	0.0903839	0.916376	3.18643e-07	2.29423e-05	CASB higher
rf	precision	casb	adasyn	40	0.0480678	0.809388	1.77032e-05	0.00113301	CASB higher
rf	precision	casb_relaxed	adasyn	40	0.0304337	0.673414	0.00029647	0.0174917	CASB higher
rf	precision	casb_edited	smote	44	-0.0406781	-0.882883	3.8563e-06	0.000258372	CASB lower
rf	precision	casb_edited	bsmote	44	-0.0301873	-0.673016	0.000515757	0.0293981	CASB lower
rf	precision	casb_edited	ros	44	-0.054319	-0.789744	1.72238e-05	0.00111955	CASB lower
rf	precision	casb_ros	bsmote	44	0.0344157	0.665165	0.000501414	0.029082	CASB higher
rf	precision	casb_ros	adasyn	40	0.0550088	0.73542	9.62702e-05	0.00577621	CASB higher
rf	recall	casb	original	44	0.124912	0.968293	9.46025e-08	7.00058e-06	CASB higher
rf	recall	casb	adasyn	40	-0.0257797	-0.684492	0.000601741	0.0336975	CASB lower
rf	recall	casb_relaxed	original	44	0.1401	0.973171	8.17884e-08	6.21592e-06	CASB higher
rf	recall	casb_edited	original	44	0.184077	0.992683	4.48182e-08	3.58545e-06	CASB higher
rf	recall	casb_edited	smote	44	0.0414626	0.762483	4.17468e-05	0.00254656	CASB higher
rf	recall	casb_edited	bsmote	44	0.0640882	0.901282	9.31423e-07	6.6131e-05	CASB higher
rf	recall	casb_edited	ros	44	0.0610241	0.79359	1.56463e-05	0.00103266	CASB higher
rf	recall	casb_ros	original	44	0.125479	0.987805	5.20696e-08	4.06143e-06	CASB higher
lr	gmean	casb	original	44	0.213706	0.969803	6.29844e-08	4.78681e-06	CASB higher
lr	gmean	casb_relaxed	original	44	0.217066	0.974448	5.44803e-08	4.19498e-06	CASB higher
lr	gmean	casb_edited	original	44	0.2232	0.962834	7.81972e-08	5.8048e-06	CASB higher
lr	gmean	casb_ros	original	44	0.216207	0.979094	4.70937e-08	3.7675e-06	CASB higher
lr	f1	casb	original	44	0.0874706	0.679443	0.000150453	0.0105317	CASB higher
lr	f1	casb_relaxed	original	44	0.088477	0.660859	0.000227223	0.0154511	CASB higher
lr	f1	casb_ros	original	44	0.0885108	0.686411	0.000128563	0.00912798	CASB higher
lr	precision	casb	adasyn	40	0.0286817	0.638462	0.000511164	0.0332257	CASB higher
lr	precision	casb_edited	original	44	-0.114668	-0.635308	0.000393939	0.026	CASB lower
lr	precision	casb_edited	smote	44	-0.017398	-0.617949	0.000770572	0.0477755	CASB lower
lr	precision	casb_edited	ros	44	-0.0136413	-0.660976	0.000269907	0.0180838	CASB lower

classifier	metric	method	baseline	n	mean_difference	rank_biserial	p_raw	p_holm	direction
lr	recall	casb	original	44	0.273321	1	5.2553e-08	4.15014e-06	CASB higher
lr	recall	casb	adasyn	40	-0.0276467	-0.821849	2.90992e-05	0.00209515	CASB lower
lr	recall	casb_relaxed	original	44	0.284644	1	7.73973e-08	5.8048e-06	CASB higher
lr	recall	casb_relaxed	adasyn	40	-0.0151917	-0.68984	0.000544666	0.0348586	CASB lower
lr	recall	casb_edited	original	44	0.309846	1	7.73973e-08	5.8048e-06	CASB higher
lr	recall	casb_edited	smote	44	0.0206549	0.728736	0.00060518	0.0381263	CASB higher
lr	recall	casb_edited	bsmote	44	0.0337878	0.734454	0.000186972	0.0129011	CASB higher
lr	recall	casb_ros	original	44	0.278774	1	5.25334e-08	4.15014e-06	CASB higher
gnb	gmean	casb_relaxed	original	44	0.0352257	0.612079	0.000638927	0.0491974	CASB higher
gnb	gmean	casb_edited	original	44	0.0462856	0.712474	4.71629e-05	0.00377303	CASB higher
gnb	gmean	casb_edited	ros	44	0.0436017	0.650055	0.000242716	0.0189319	CASB higher
gnb	gmean	casb_ros	smote	44	-0.0306772	-0.705128	0.00012423	0.00981419	CASB lower

CASB-edited against each baseline

classifier	metric	baseline	n	mean_difference	p_holm	direction
svm	gmean	original	44	0.213274	3.05842e-06	CASB higher
svm	gmean	smote	44	0.00925345	1	not significant
svm	gmean	bsmote	44	0.0271381	1	not significant
svm	gmean	ros	44	0.000549843	1	not significant
svm	gmean	adasyn	40	0.0113946	1	not significant
svm	gmean	rus	44	0.0190496	1	not significant
svm	gmean	smoteenn	44	-0.00463344	1	not significant
svm	f1	original	44	0.101802	0.00647089	CASB higher
svm	f1	smote	44	-0.00651806	1	not significant
svm	f1	bsmote	44	0.00219625	1	not significant
svm	f1	ros	44	-0.00328261	1	not significant
svm	f1	adasyn	40	0.0136569	1	not significant
svm	f1	rus	44	0.04651	0.0220534	CASB higher
svm	f1	smoteenn	44	0.00993323	1	not significant
svm	precision	original	44	-0.0527055	0.947555	not significant
svm	precision	smote	44	-0.0228569	0.00133931	CASB lower
svm	precision	bsmote	44	-0.0067501	0.61089	not significant
svm	precision	ros	44	-0.0144598	0.0470612	CASB lower
svm	precision	adasyn	40	0.0121785	1	not significant

classifier	metric	baseline	n	mean_difference	p_holm	direction
svm	precision	rus	44	0.0482942	0.42997	not significant
svm	precision	smoteenn	44	0.00884858	1	not significant
svm	recall	original	44	0.262814	3.95922e-06	CASB higher
svm	recall	smote	44	0.0252893	0.000214143	CASB higher
svm	recall	bsmote	44	0.0409977	0.0105256	CASB higher
svm	recall	ros	44	0.0146654	0.42997	not significant
svm	recall	adasyn	40	0.0211264	1	not significant
svm	recall	rus	44	0.00862595	1	not significant
svm	recall	smoteenn	44	-0.00731591	1	not significant
knn	gmean	original	44	0.159353	1.95901e-05	CASB higher
knn	gmean	smote	44	0.00618646	1	not significant
knn	gmean	bsmote	44	0.0134921	1	not significant
knn	gmean	ros	44	0.0154212	0.2708	not significant
knn	gmean	adasyn	40	0.00204011	1	not significant
knn	gmean	rus	44	0.0115044	1	not significant
knn	gmean	smoteenn	44	0.00126225	1	not significant
knn	f1	original	44	0.0215531	1	not significant
knn	f1	smote	44	-0.0114998	1	not significant
knn	f1	bsmote	44	-0.0148175	1	not significant
knn	f1	ros	44	-0.00989208	1	not significant
knn	f1	adasyn	40	0.00251514	1	not significant
knn	f1	rus	44	0.0300866	1	not significant
knn	f1	smoteenn	44	0.00752185	0.236767	not significant
knn	precision	original	44	-0.13386	0.00187284	CASB lower
knn	precision	smote	44	-0.0278921	0.00018454	CASB lower
knn	precision	bsmote	44	-0.0228286	1	not significant
knn	precision	ros	44	-0.0237374	0.00606729	CASB lower
knn	precision	adasyn	40	0.00125706	1	not significant
knn	precision	rus	44	0.0215138	1	not significant
knn	precision	smoteenn	44	0.00604997	1	not significant
knn	recall	original	44	0.221753	5.87937e-06	CASB higher
knn	recall	smote	44	0.0254738	0.00957531	CASB higher
knn	recall	bsmote	44	0.0295857	0.27513	not significant
knn	recall	ros	44	0.0399743	0.00184207	CASB higher
knn	recall	adasyn	40	0.00617384	1	not significant

classifier	metric	baseline	n	mean_difference	p_holm	direction
knn	recall	rus	44	0.00759296	1	not significant
knn	recall	smoteenn	44	-0.00833944	1	not significant
rf	gmean	original	44	0.148854	4.19498e-06	CASB higher
rf	gmean	smote	44	0.0197218	0.132084	not significant
rf	gmean	bsmote	44	0.0485127	0.00253105	CASB higher
rf	gmean	ros	44	0.0393673	0.00246007	CASB higher
rf	gmean	adasyn	40	0.0203103	1	not significant
rf	f1	original	44	0.0883484	0.000190888	CASB higher
rf	f1	smote	44	-0.00208435	1	not significant
rf	f1	bsmote	44	0.0167933	0.928261	not significant
rf	f1	ros	44	0.00252229	1	not significant
rf	f1	adasyn	40	0.0079158	1	not significant
rf	precision	original	44	-0.0400566	0.646125	not significant
rf	precision	smote	44	-0.0406781	0.000258372	CASB lower
rf	precision	bsmote	44	-0.0301873	0.0293981	CASB lower
rf	precision	ros	44	-0.054319	0.00111955	CASB lower
rf	precision	adasyn	40	-0.0160546	0.667477	not significant
rf	recall	original	44	0.184077	3.58545e-06	CASB higher
rf	recall	smote	44	0.0414626	0.00254656	CASB higher
rf	recall	bsmote	44	0.0640882	6.6131e-05	CASB higher
rf	recall	ros	44	0.0610241	0.00103266	CASB higher
rf	recall	adasyn	40	0.0393016	0.12138	not significant
lr	gmean	original	44	0.2232	5.8048e-06	CASB higher
lr	gmean	smote	44	0.00309773	1	not significant
lr	gmean	bsmote	44	0.0157538	1	not significant
lr	gmean	ros	44	0.00163007	1	not significant
lr	gmean	adasyn	40	0.00200115	1	not significant
lr	f1	original	44	0.0810163	0.169499	not significant
lr	f1	smote	44	-0.0050077	1	not significant
lr	f1	bsmote	44	-0.00075586	1	not significant
lr	f1	ros	44	-0.00103115	1	not significant
lr	f1	adasyn	40	0.00327475	1	not significant
lr	precision	original	44	-0.114668	0.026	CASB lower
lr	precision	smote	44	-0.017398	0.0477755	CASB lower
lr	precision	bsmote	44	-0.00812419	1	not significant

classifier	metric	baseline	n	mean_difference	p_holm	direction
lr	precision	ros	44	-0.0136413	0.0180838	CASB lower
lr	precision	adasyn	40	-0.00026104	1	not significant
lr	recall	original	44	0.309846	5.8048e-06	CASB higher
lr	recall	smote	44	0.0206549	0.0381263	CASB higher
lr	recall	bsmote	44	0.0337878	0.0129011	CASB higher
lr	recall	ros	44	0.0169393	0.597793	not significant
lr	recall	adasyn	40	0.0125301	1	not significant
gnb	gmean	original	44	0.0462856	0.00377303	CASB higher
gnb	gmean	smote	44	0.00620431	1	not significant
gnb	gmean	bsmote	44	0.00772867	1	not significant
gnb	gmean	ros	44	0.0436017	0.0189319	CASB higher
gnb	gmean	adasyn	40	0.018725	1	not significant
gnb	f1	original	44	0.02845	0.351094	not significant
gnb	f1	smote	44	0.0142659	1	not significant
gnb	f1	bsmote	44	0.0142211	1	not significant
gnb	f1	ros	44	0.034415	0.232819	not significant
gnb	f1	adasyn	40	0.0167876	1	not significant
gnb	precision	original	44	0.0108225	1	not significant
gnb	precision	smote	44	0.00992117	1	not significant
gnb	precision	bsmote	44	0.0143746	1	not significant
gnb	precision	ros	44	0.0294813	1	not significant
gnb	precision	adasyn	40	0.016334	1	not significant
gnb	recall	original	44	0.0342914	1	not significant
gnb	recall	smote	44	-0.00230856	1	not significant
gnb	recall	bsmote	44	0.00908318	1	not significant
gnb	recall	ros	44	-0.00230309	1	not significant
gnb	recall	adasyn	40	-0.0157349	1	not significant

E. Family-level performance

The following totals include all compatible baselines, including no resampling and, for SVM/KNN, RUS and SMOTE-ENN. The CSV also reports the original four-resampler population separately. A family win selects the best variant retrospectively; it measures an upper envelope of observed scores rather than expected deployable selector performance. No significance test is applied to this oracle. Joint wins require the same variant to beat the best baseline separately on both G-mean and F1.

classifier	metric	n	wins	ties	losses	win_percent	majority
gnb	f1	40.000	10.000	0.000	30.000	25.000	False
gnb	gmean	40.000	12.000	0.000	28.000	30.000	False

classifier	metric	n	wins	ties	losses	win_percent	majority
gnb	joint_gmean_f1_same_variant	40.000	10.000	0.000	30.000	25.000	False
gnb	precision	40.000	8.000	1.000	31.000	20.000	False
gnb	recall	40.000	10.000	15.000	15.000	25.000	False
knn	f1	40.000	14.000	2.000	24.000	35.000	False
knn	gmean	40.000	11.000	1.000	28.000	27.500	False
knn	joint_gmean_f1_same_variant	40.000	8.000	1.000	31.000	20.000	False
knn	precision	40.000	5.000	1.000	34.000	12.500	False
knn	recall	40.000	5.000	4.000	31.000	12.500	False
lr	f1	40.000	15.000	3.000	22.000	37.500	False
lr	gmean	40.000	15.000	2.000	23.000	37.500	False
lr	joint_gmean_f1_same_variant	40.000	9.000	2.000	29.000	22.500	False
lr	precision	40.000	4.000	1.000	35.000	10.000	False
lr	recall	40.000	16.000	7.000	17.000	40.000	False
rf	f1	40.000	26.000	2.000	12.000	65.000	True
rf	gmean	40.000	27.000	1.000	12.000	67.500	True
rf	joint_gmean_f1_same_variant	40.000	19.000	2.000	19.000	47.500	False
rf	precision	40.000	11.000	4.000	25.000	27.500	False
rf	recall	40.000	25.000	7.000	8.000	62.500	True
svm	f1	40.000	17.000	3.000	20.000	42.500	False
svm	gmean	40.000	9.000	3.000	28.000	22.500	False
svm	joint_gmean_f1_same_variant	40.000	6.000	3.000	31.000	15.000	False
svm	precision	40.000	8.000	3.000	29.000	20.000	False
svm	recall	40.000	5.000	7.000	28.000	12.500	False

Dataset dependence and sensitivity

Named source families are assigned by dataset identity (Abalone, Ecoli, Glass, Haberman, Iris, New-thyroid, Page-blocks, Pima, Segment, Shuttle, Vehicle, Vowel, Wisconsin, Yeast). Two sensitivity views are prespecified: equal means of valid paired differences within each source family; and one lexicographically first named task per source family, chosen without consulting performance. A representative task with a failed method remains missing for that pair and is not replaced with a favourable task. Exact sign tests are also Holm-corrected. These analyses reduce over-weighting of related variants but cannot prove family independence.

source_family_mean

classifier	not significant
gnb	80
knn	112

classifier	not significant
lr	80
rf	80
svm	112

Corrected exact sign-test rejections: 9.

one_task_per_family

classifier	not significant
gnb	80
knn	112
lr	80
rf	80
svm	112

Corrected exact sign-test rejections: 0.

F. Discussion

CASB-edited can trade precision for recall because its unrestricted generation and majority-only ENN alter the training geometry and class balance. These are plausible interpretations of the implementation, not a demonstrated causal explanation. A matched generation/cleaning ablation with identical quotas, ratios and classifier tuning is needed to attribute a gain to a particular component. A higher collection mean can coexist with non-significant pairwise results or a minority of dataset wins.

G. Limitations and discrepancies from earlier summaries

Earlier averages compared the CASB family with four resamplers on 40 complete datasets. This report adds the no-resampling reference and compatible RUS/SMOTE-ENN baselines and uses 44 valid pairs where possible. Stronger baseline maxima can reduce family wins. Holm correction now has 112 or 80 hypotheses within classifier rather than a global 192/320-view family; exact p-values therefore need not match older reports. No pooled classifier view is a primary test. The 80-scope column permits an explicit scope comparison.

Non-significant differences do not establish equivalence. G-mean/F1 consistency, precision-recall trade-offs, ADASYN failures, related KEEL variants, reused development datasets, retrospective variant/classifier selection, absent independent confirmation and unmatched cleaning ablations restrict claims. GaussianNB has the same stored protocol but is a separate additional classifier analysis. Holm control is within each declared family, not across every result in the report. Friedman uses the asymptotic chi-square approximation; the complete cohorts exceed ten datasets and compare more than six methods.

H. Future work and paper-ready conclusion

All four classifiers have significant Friedman differences for each metric after correcting four omnibus tests. Against all available baselines, RF alone has majority family wins for both G-mean (27/40) and F1 (26/40); its same-variant joint count is 19/40, below a majority. Against only the original four resamplers, RF wins 28/40 for G-mean, 26/40 for F1, and 20/40 jointly (exactly half). These are not counts out of 44.

CASB-edited with RF has corrected G-mean gains over Borderline-SMOTE and ROS and corrected recall gains over SMOTE, Borderline-SMOTE and ROS. It also has significant precision losses against those three methods. No fixed-variant Wilcoxon comparison survives either source-family sensitivity analysis after Holm correction.

The experiments support describing CASB as an evaluated SMOTE-based family with classifier- and metric-dependent gains and losses. They do not support universal superiority or establish novelty. A paper should present all corrected beneficial and harmful comparisons, disclose the post hoc family envelope and classifier selection, and distinguish performance from mechanism. Future work should test independent datasets, prespecify variant selection using training data, compare matched cleaning ablations, and improve G-mean/F1 consistency without claiming these changes are already achieved.

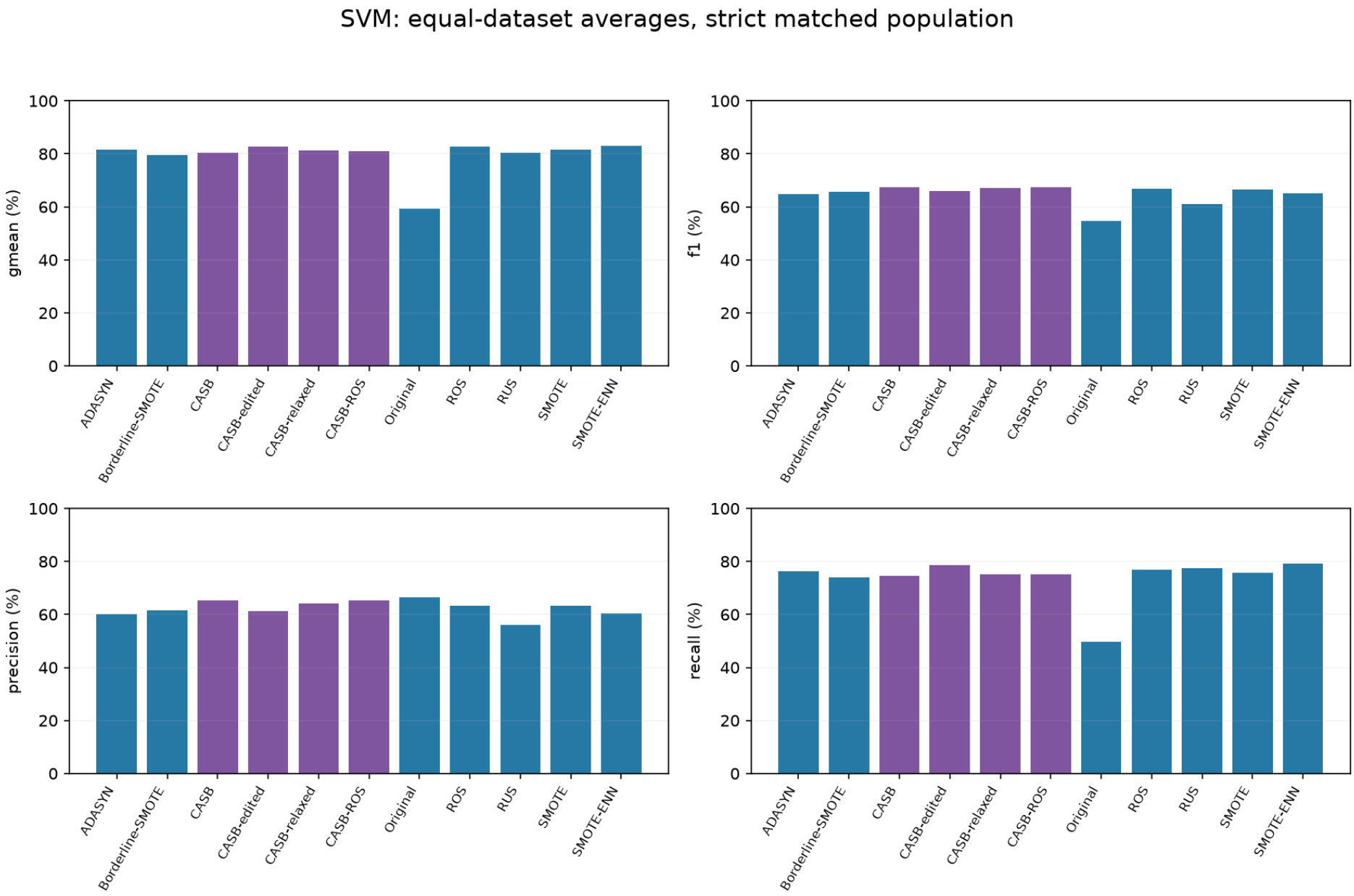
Verification and reproducibility

Confusion-matrix metrics and training-only ratio choices were verified. Original notebook records were deduplicated by dataset/classifier/method/seed/fold. All 440 train/test file hashes matched notebook provenance. CSV means were cross-checked with independent arithmetic aggregation; Holm monotonicity, all-tie handling, paired-unit counts, Friedman ranks and family-win denominators are asserted by the analysis. Existing notebooks, code, progress report and datasets are SHA256-checked before/after analysis; hashes are embedded in the executed notebook. No training is imported or executed.

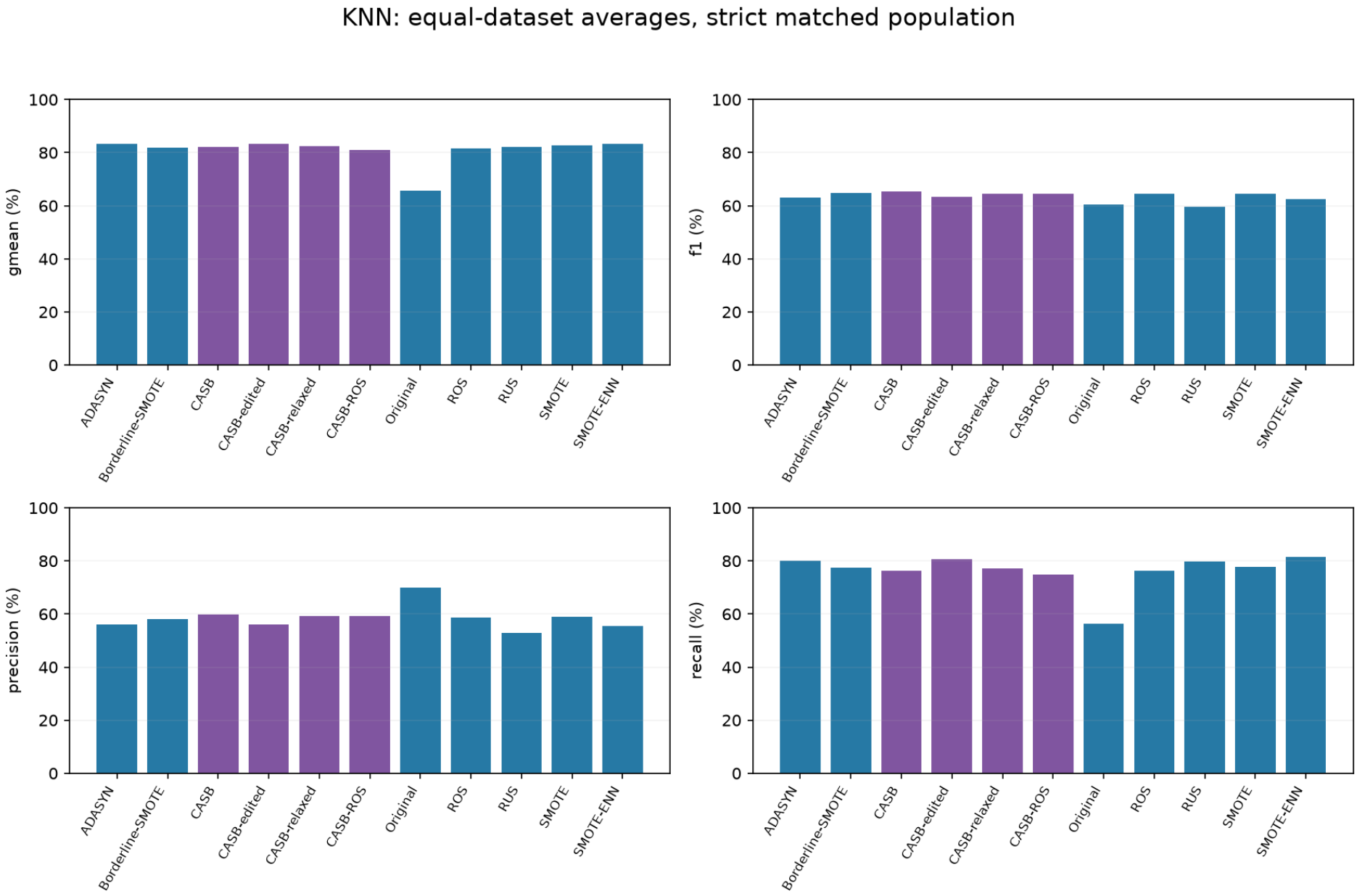
Runtime: {"numpy": "2.5.4", "pandas": "3.0.6", "scipy": "1.18.1", "matplotlib": "3.11.2"}.

Methods references: [Demsar (2006)](https://jmlr.org/papers/v7/demsar06a.html); [SciPy Wilcoxon](https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.wilcoxon.html); [SciPy Friedman](https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.friedmanchisquare.html).

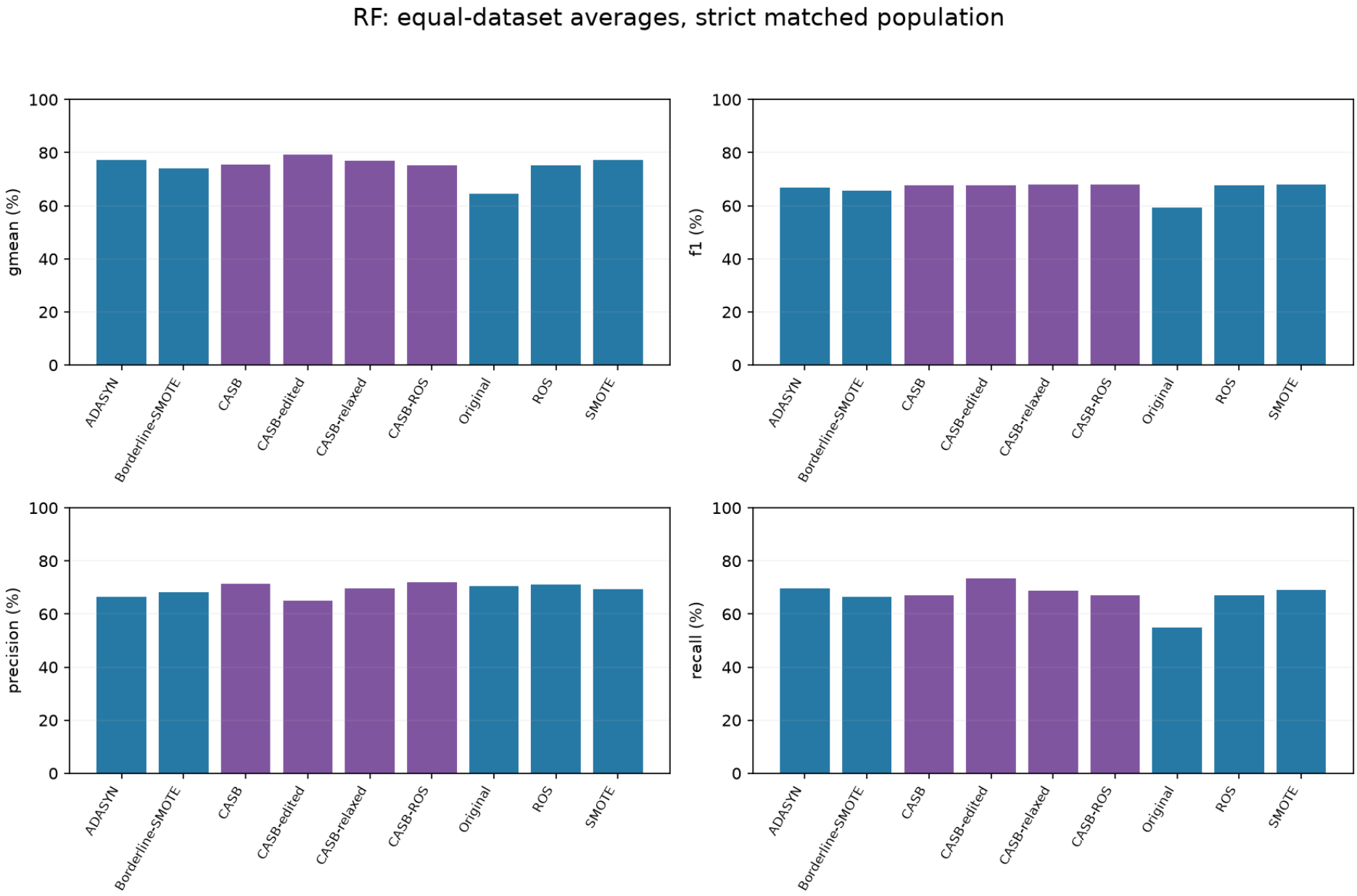
SVM: equal-dataset averages, strict matched population



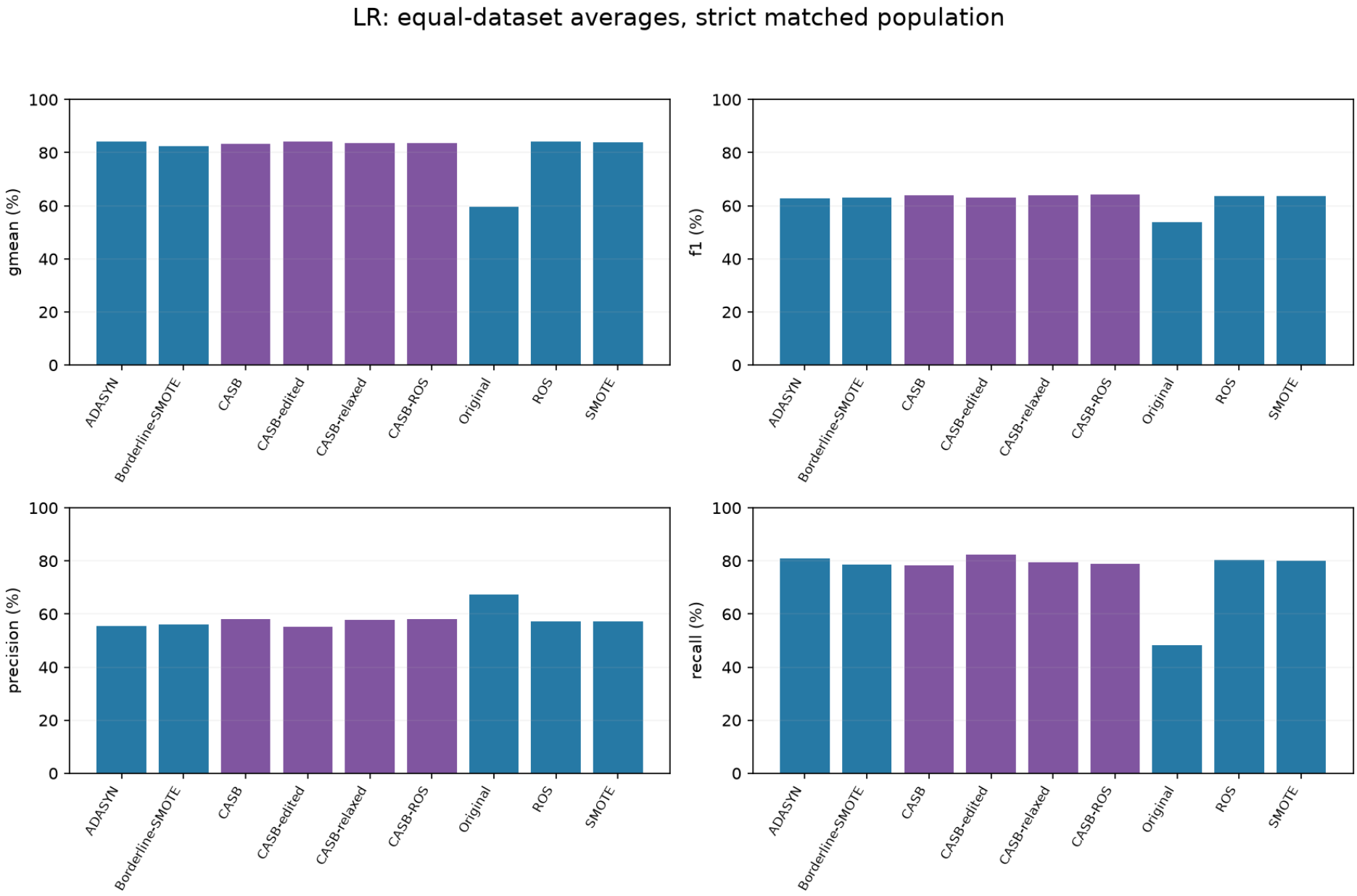
KNN: equal-dataset averages, strict matched population



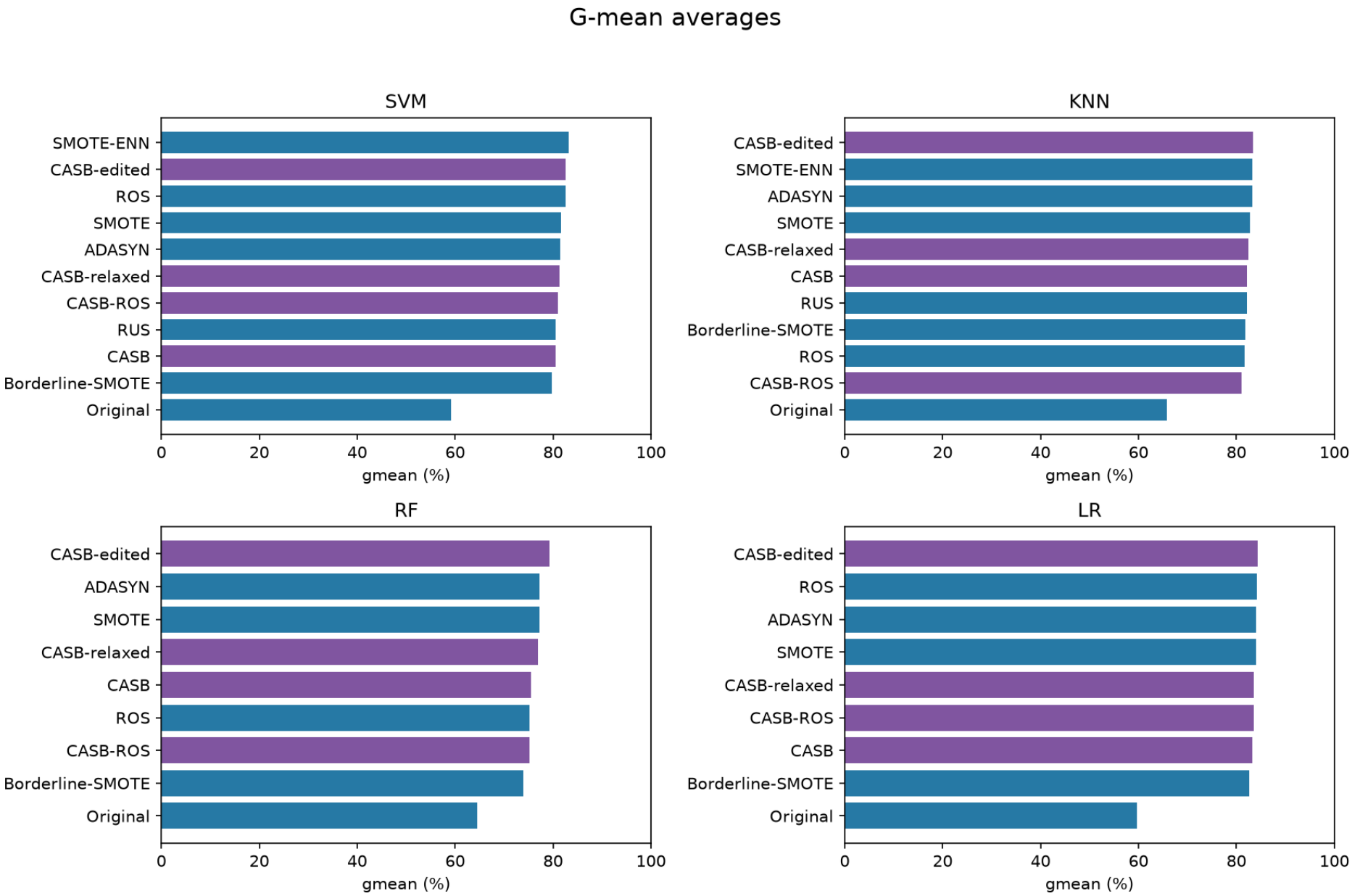
RF: equal-dataset averages, strict matched population



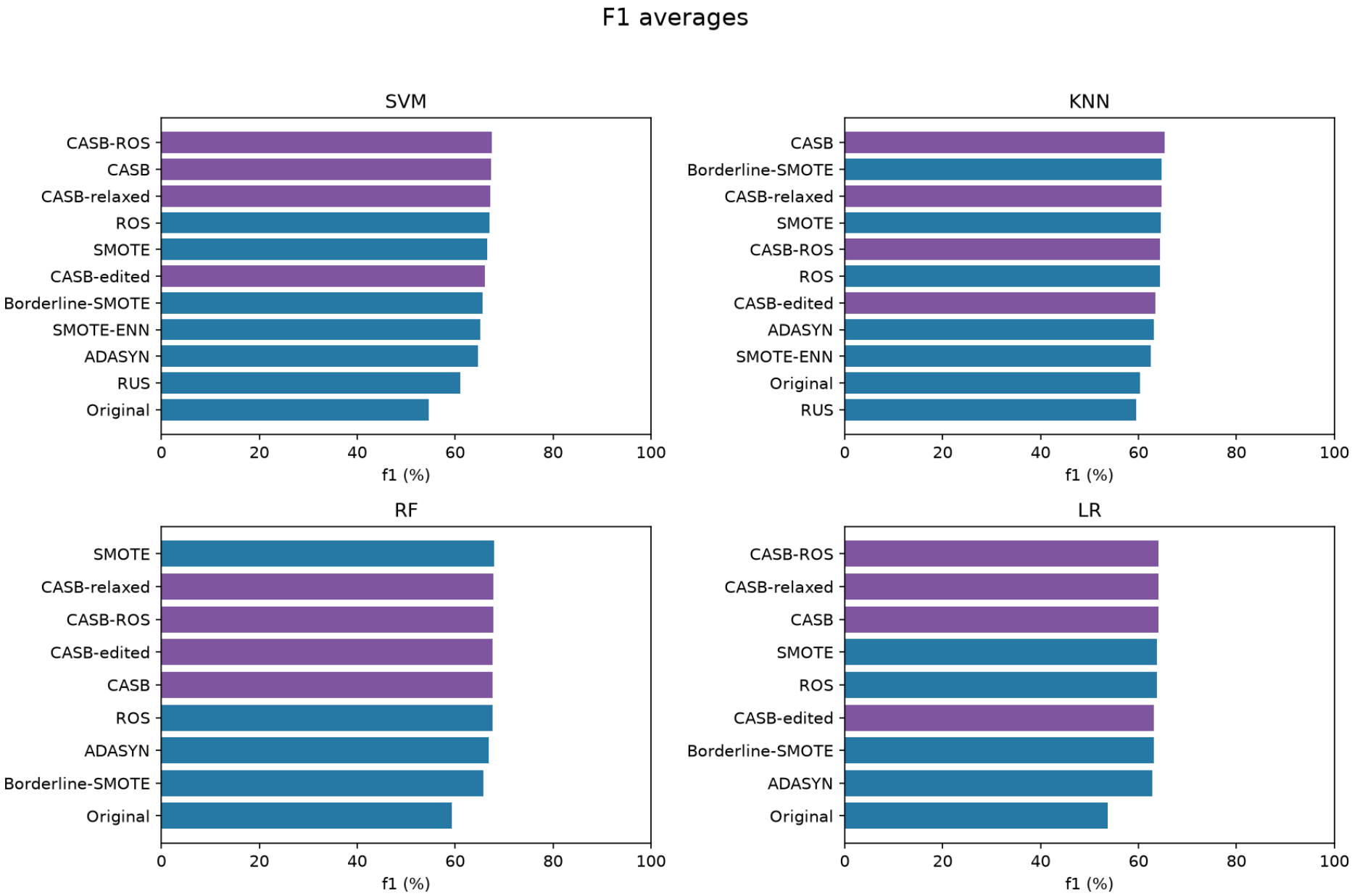
LR: equal-dataset averages, strict matched population



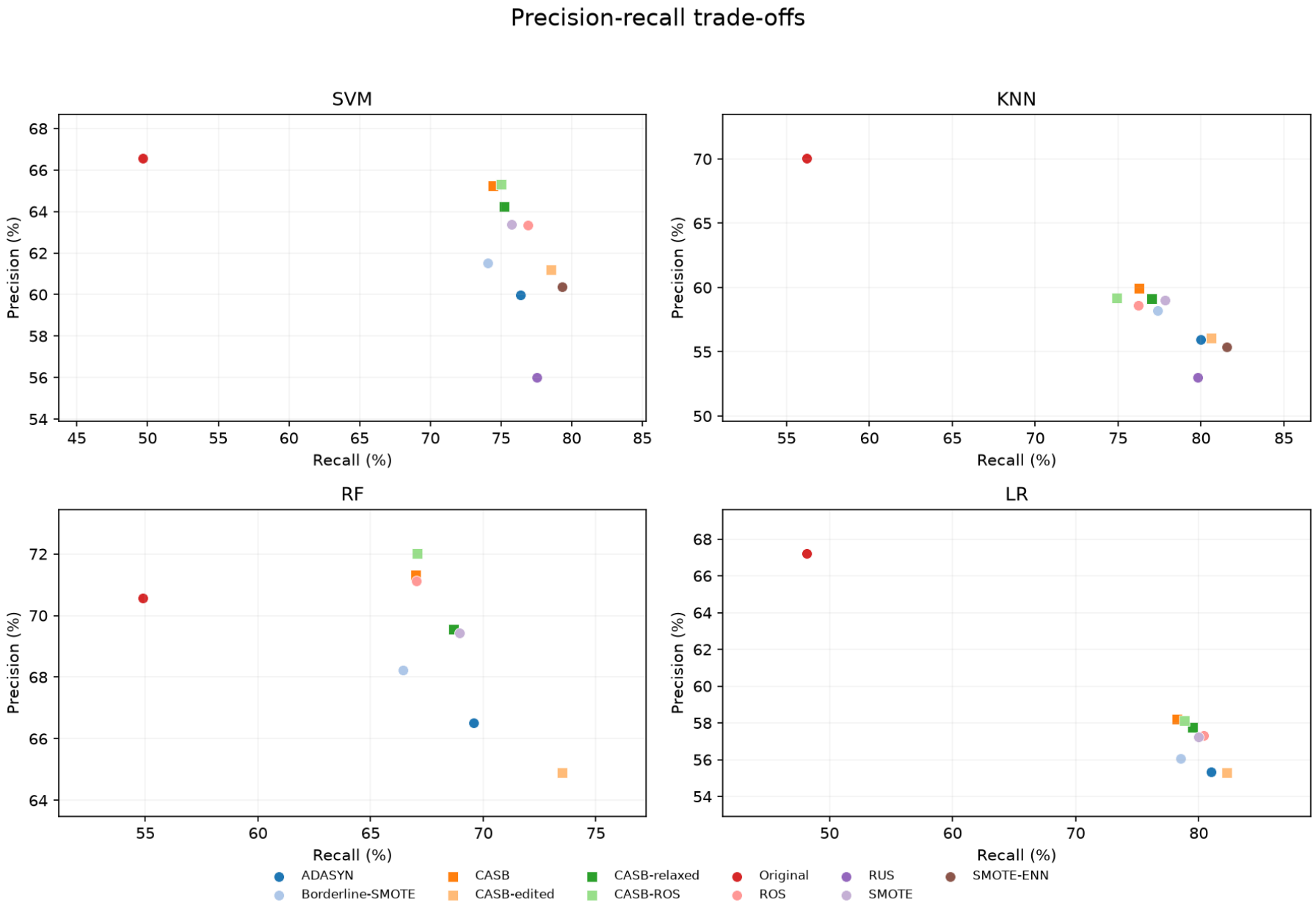
G-mean averages



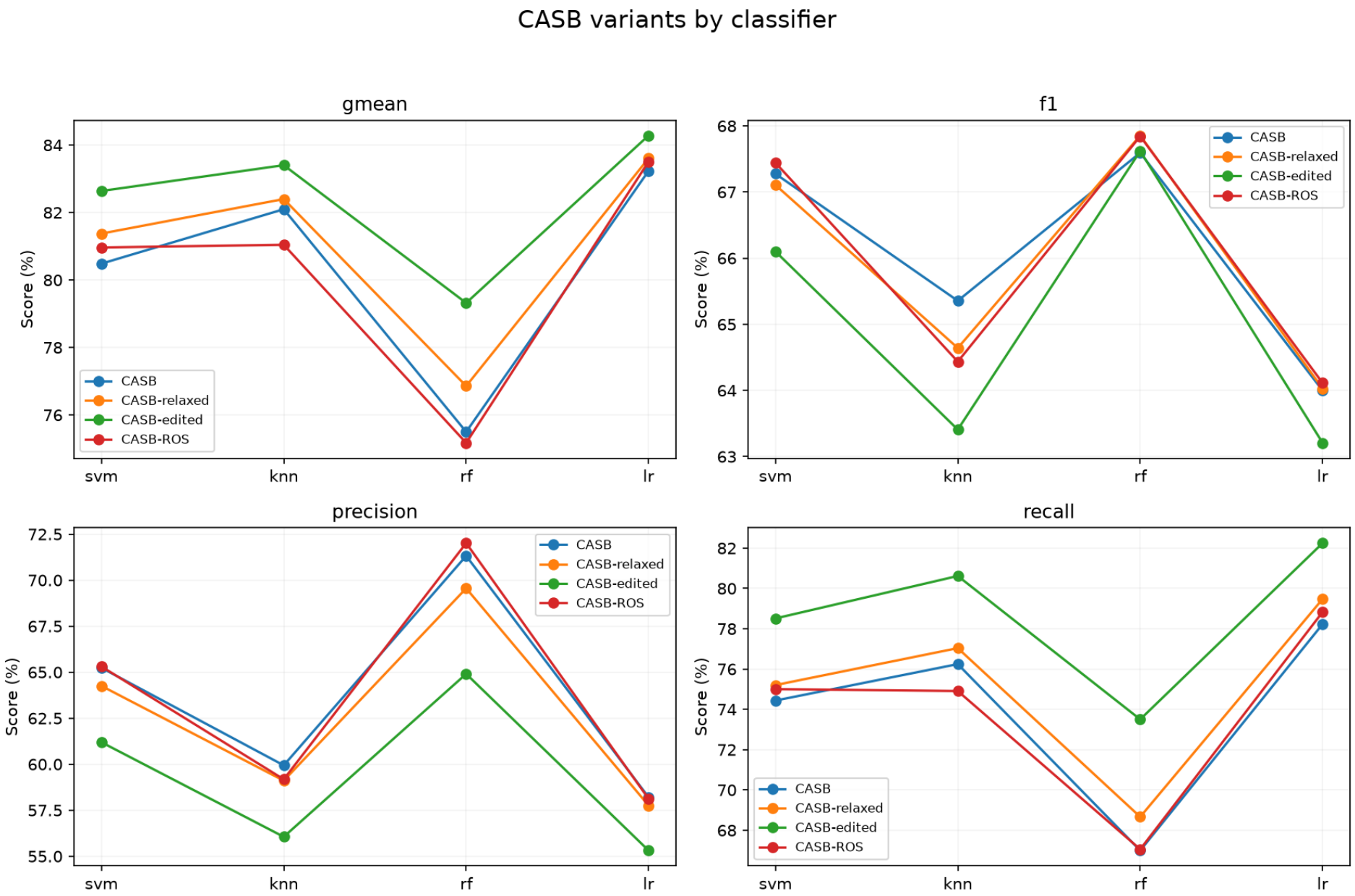
F1 averages


F1 averages
SVM
CASB-ROS
CASB
CASB-relaxed
ROS
SMOTE
CASB-edited
Borderline-SMOTE
SMOTE-ENN
ADASYN
RUS
Original
0
20
40
60
80
100
f1 (%)
KNN
CASB
Borderline-SMOTE
CASB-relaxed
SMOTE
CASB-ROS
ROS
CASB-edited
ADASYN
SMOTE-ENN
Original
RUS
0
20
40
60
80
100
f1 (%)
RF
SMOTE
CASB-relaxed
CASB-ROS
CASB-edited
CASB
ROS
ADASYN
Borderline-SMOTE
Original
0
20
40
60
80
100
f1 (%)
LR
CASB-ROS
CASB-relaxed
CASB
SMOTE
ROS
CASB-edited
Borderline-SMOTE
ADASYN
Original
0
20
40
60
80
100
f1 (%)

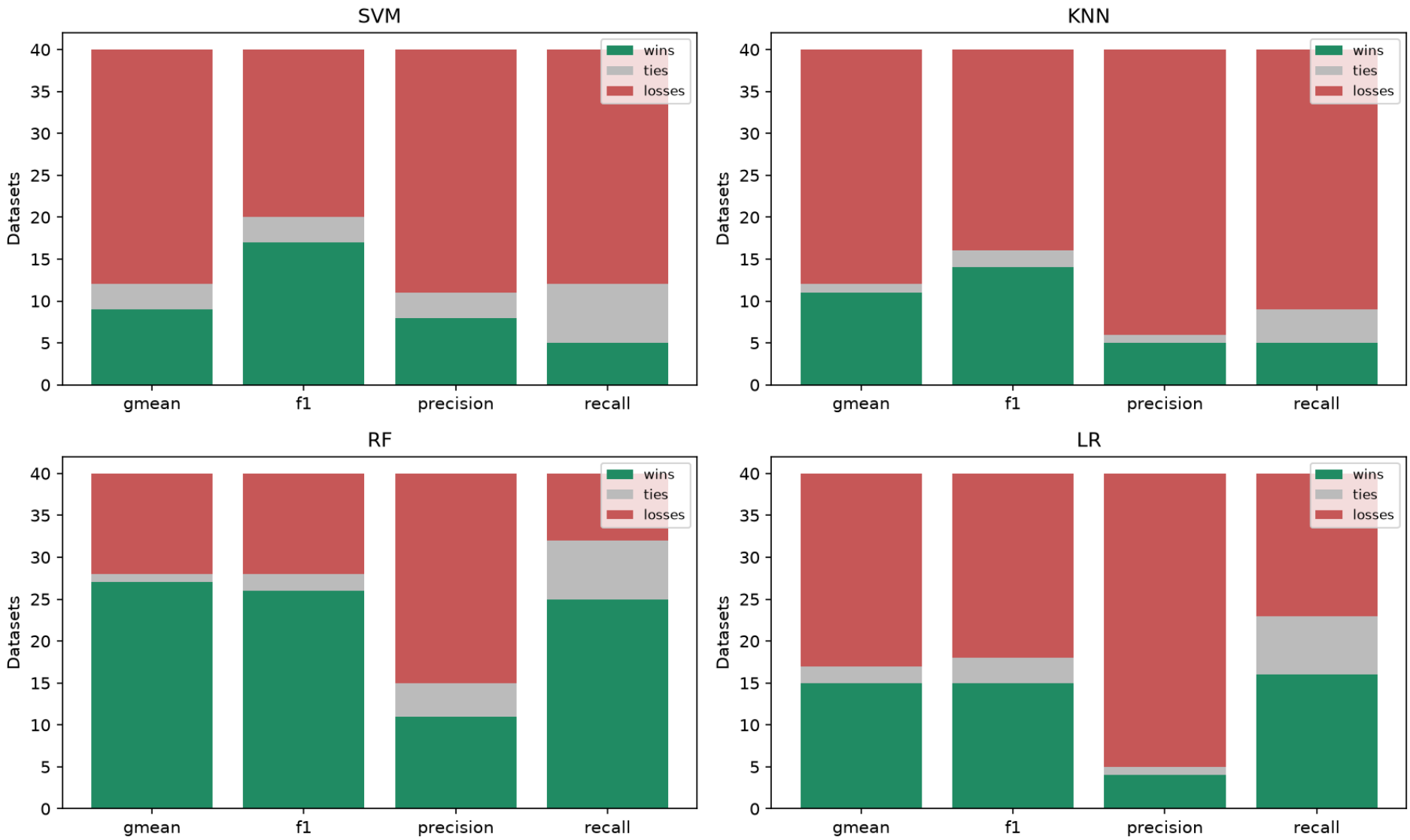
Precision-recall trade-offs



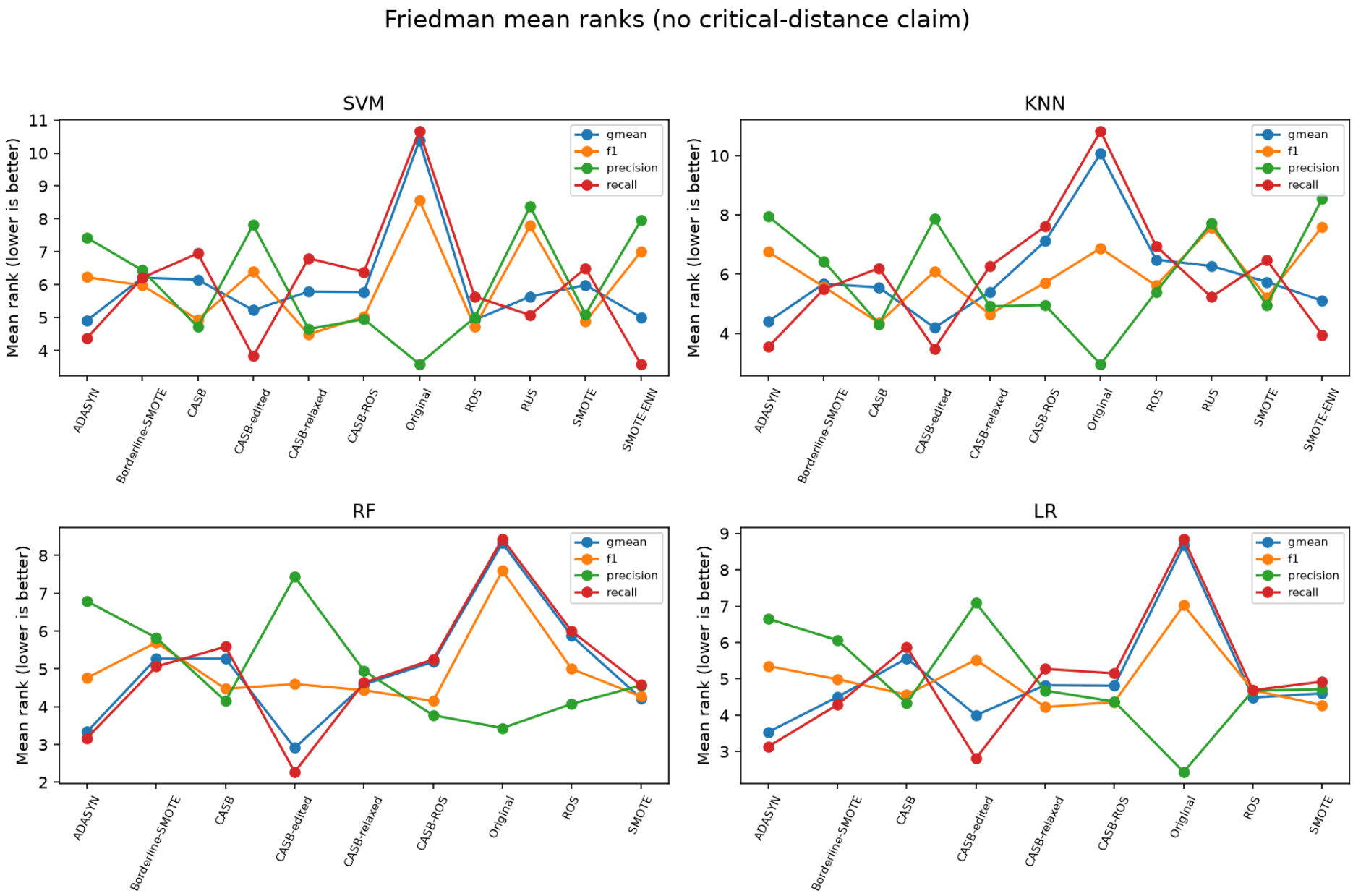
CASB variants by classifier



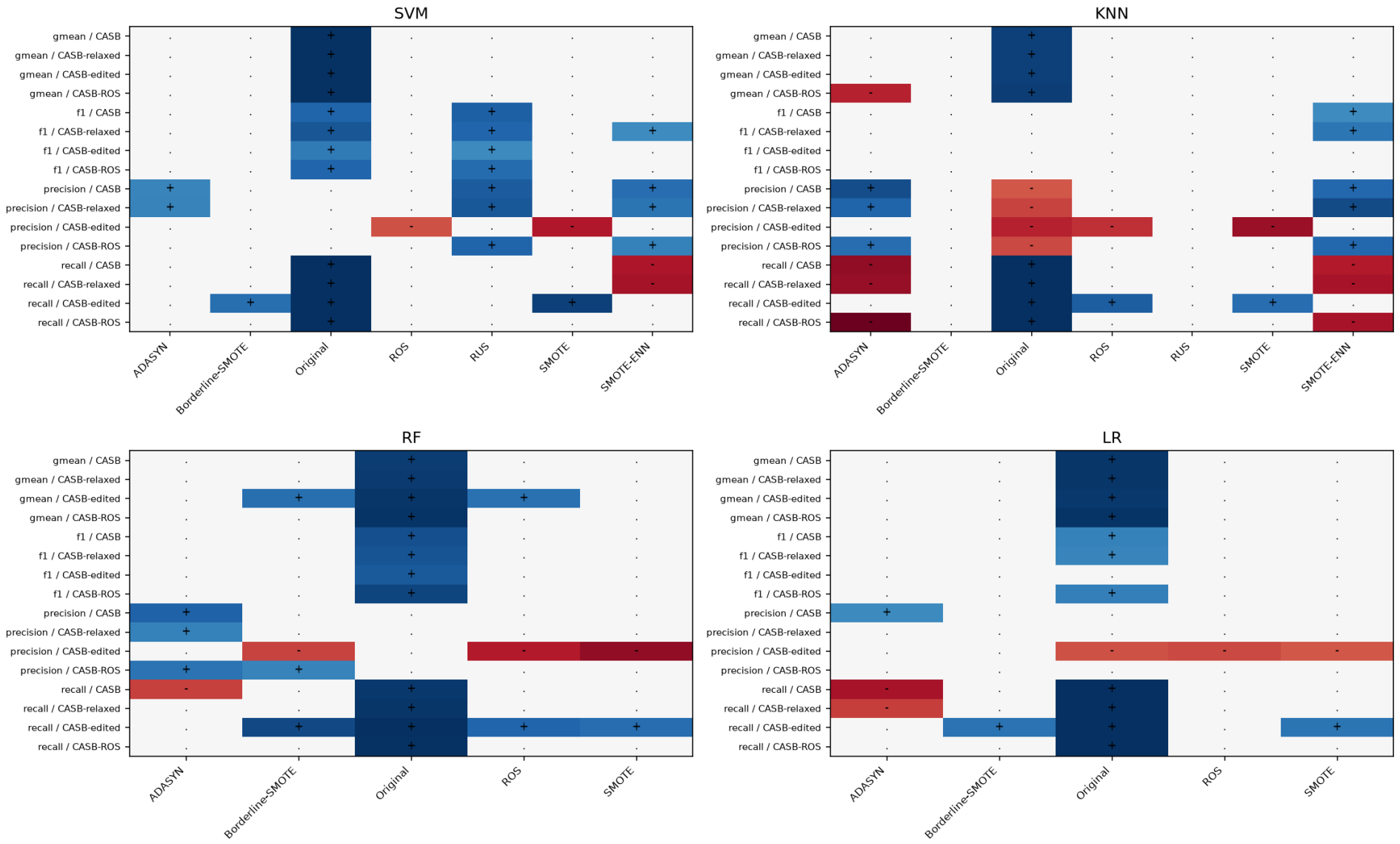
Retrospective family win / tie / loss against all available baselines



Friedman mean ranks (no critical-distance claim)



Wilcoxon-Holm: + significant CASB benefit, - significant harm, . non-significant



Family win rates against all available baselines (retrospective)

